A tutorial:
Materials informatics for molecules
…a.k.a. chemoinformatics

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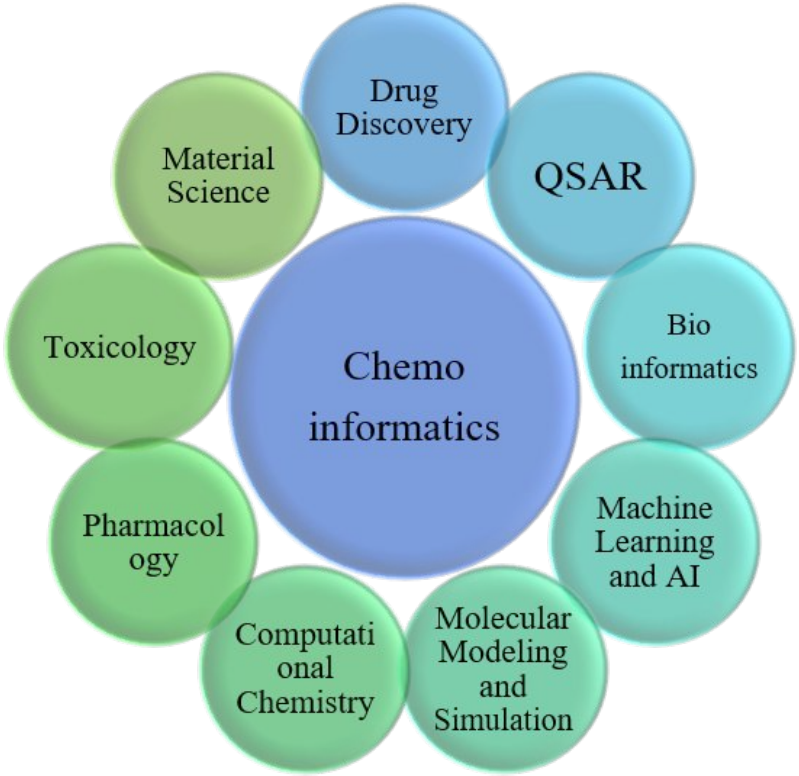
19.03.2026

Introduction. What is chemoinformatics?



- the set of computer algorithms and tools to store and analyse chemical data
- mostly it appears in context of drug design, but the application field is widening thanks to ML & AI

If you're not familiar with materials informatics, please consider "Introduction to Materials Informatics" course in Term 1b

PubChem: an experimental database for molecules

National Library of Medicine
National Center for Biotechnology Information

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COMPOUND

Benzene

PubChem CID	241
Structure	2D, 3D, Crystal
Primary Hazards	Flammable, Irritant, Health Hazard Laboratory Chemical Safety Summary (LCSS) Datasheet
Molecular Formula	C_6H_6

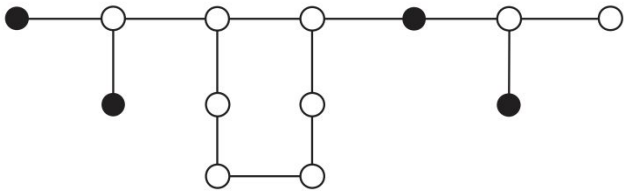
Cite Download

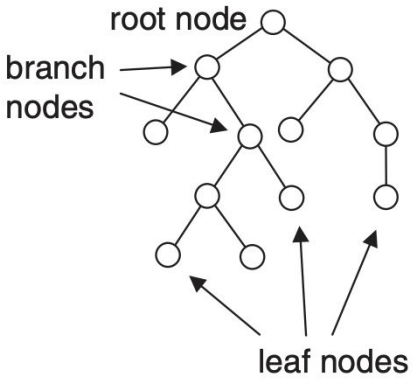
CONTENTS

Title and Summary
1 Structures
2 Names and Identifiers
3 Chemical and Physical Properties
4 Spectral Information
5 Related Records
6 Chemical Vendors
7 Drug and Medication Information
8 Food Additives and Ingredients
9 Pharmacology and Biochemistry
10 Use and Manufacturing
11 Identification
12 Safety and Hazards
13 Toxicity
14 Associated Disorders and Diseases

Benzene example: https://pubchem.ncbi.nlm.nih.gov/compound/Benzene#section=InChI

SMILES – a string format to represent a molecule





- graph shows connectivity
- atomic type (nodes) & bond type (edges) should be included

SMILES – a string format to represent a molecule

Option 1:
connectivity matrix file

MDL (right), PDB files

number of bonds

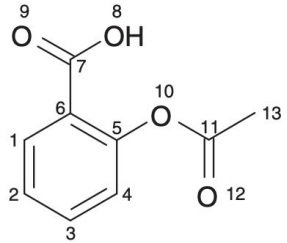
number of atoms

the first atom is a carbon

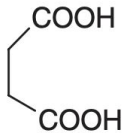
```
-ISIS-  09270222202D

13 13  0  0  0  0  0  0  0  0999 V2000
  -3.4639   -1.5375    0.0000 C  0  0  0  0  0  0  0  0  0  0  0  0
  -3.4651   -2.3648    0.0000 C  0  0  0  0  0  0  0  0  0  0  0  0
  -2.7503   -2.7777    0.0000 C  0  0  0  0  0  0  0  0  0  0  0  0
  -2.0338   -2.3644    0.0000 C  0  0  0  0  0  0  0  0  0  0  0  0
  -2.0367   -1.5338    0.0000 C  0  0  0  0  0  0  0  0  0  0  0  0
  -2.7521   -1.1247    0.0000 C  0  0  0  0  0  0  0  0  0  0  0  0
  -2.7545   -0.2997    0.0000 C  0  0  0  0  0  0  0  0  0  0  0  0
  -2.0413    0.1149    0.0000 O  0  0  0  0  0  0  0  0  0  0  0  0
  -3.4702    0.1107    0.0000 O  0  0  0  0  0  0  0  0  0  0  0  0
  -1.3238   -1.1186    0.0000 O  0  0  0  0  0  0  0  0  0  0  0  0
  -0.6125   -1.5292    0.0000 C  0  0  0  0  0  0  0  0  0  0  0  0
  -0.6167   -2.3542    0.0000 O  0  0  0  0  0  0  0  0  0  0  0  0
   0.1000   -1.1125    0.0000 C  0  0  0  0  0  0  0  0  0  0  0  0
 1  2  2  0  0  0  0
 6  7  1  0  0  0  0
 3  4  2  0  0  0  0
 7  8  1  0  0  0  0
 7  9  2  0  0  0  0
 4  5  1  0  0  0  0
 5 10  1  0  0  0  0
 2  3  1  0  0  0  0
10 11  1  0  0  0  0
 5  6  2  0  0  0  0
11 12  2  0  0  0  0
 6  1  1  0  0  0  0
11 13  1  0  0  0  0
M END
```

the first three numbers are the *x*, *y* and *z* coordinates of the atom

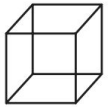
the first bond is between atoms 1 and 2 and has order 2



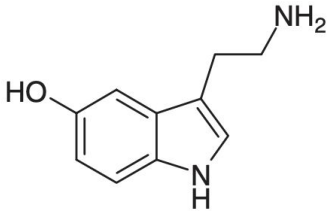
SMILES – a string format to represent a molecule



succinicacid:
OC(=O)CCC(=O)O



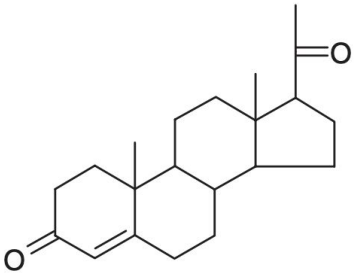
cubane:
C1(C2C3C14)C5C2C3C45



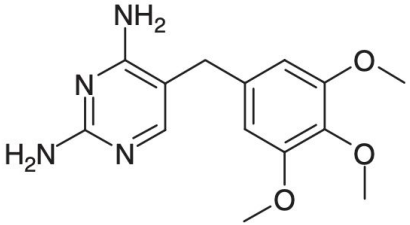
serotonin:
NCCc1c[nH]c2ccc(O)cc12

*Option 2**:
string format SMILES

*there are other string formats like InChI, but SMILES is the most common



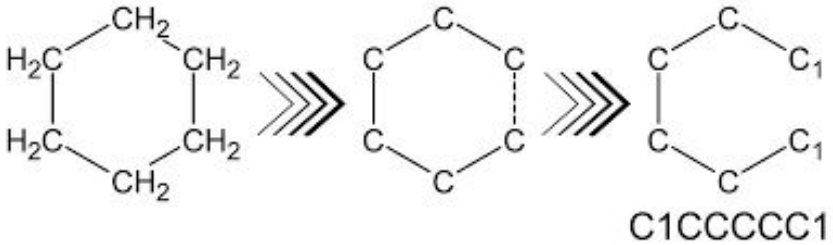
progesterone:
CC(=O)C1CCC2C3CCC4=CC(=O)CCC4(C)C3CCC12C

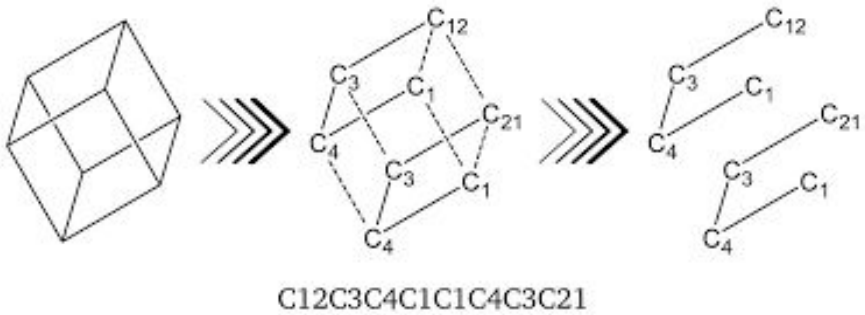


trimethoprim: COc1cc(Cc2cnc(N)nc2N)cc(OC)c1OC

SMILES – a string format to represent a molecule

How SMILES string is constructed:

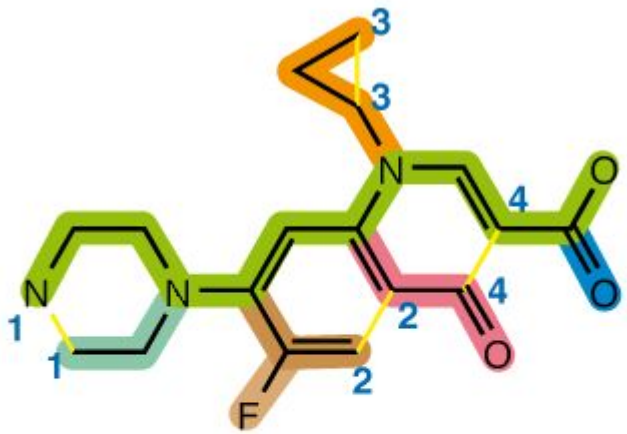




- hydrogens are assessed based on carbon valence of 4
- closing the cycles with atom numbering
- bond type:
  - C-C == CC
  - C=C == C=C
  - C≡C == C#C

SMILES – a string format to represent a molecule

How SMILES string is constructed:

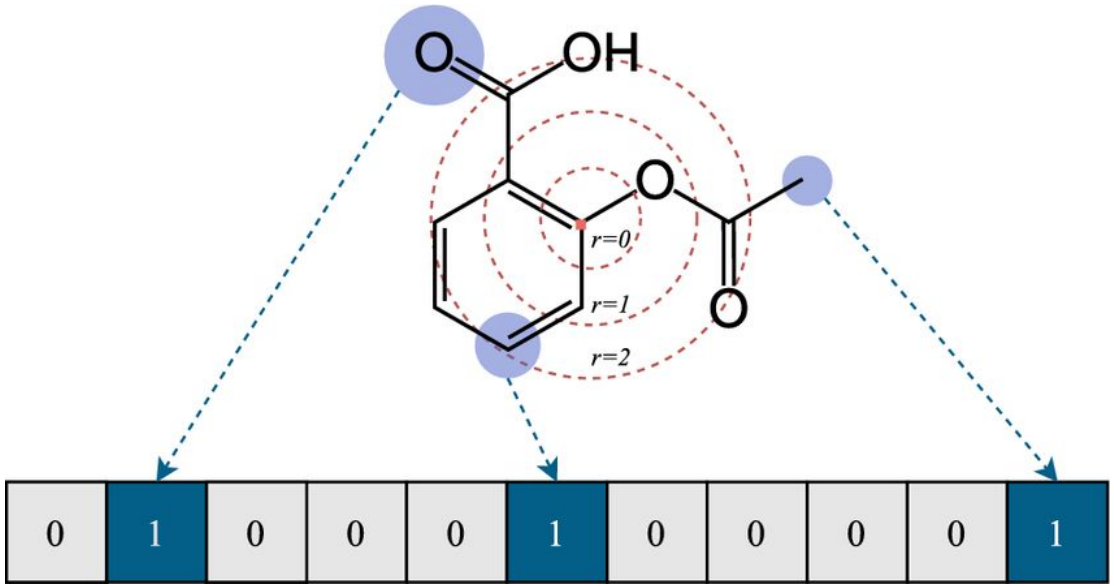


N1CCN(CC1)C(C(F)=C2)=CC(=C2C4=O)N(C3CC3)C=C4C(=O)O

Benzene example: https://pubchem.ncbi.nlm.nih.gov/compound/Benzene#section=InChI

Morgan Fingerprints (Extended Connectivity FingerPrints)

- a format of molecules representation accessible for algorithms
- determines a local surrounding for each atom in the molecule in form of binary vector



Morgan Fingerprints (Extended Connectivity FingerPrints)

1. Determine a fragment according to radius

2. Encoder algorithm creates a hash for each fragment (e.g. 9845271, non-interpretable for humans – it's like a barcode for products)

3. Divide hash number by the set number of bits (e.g. 2048) to get the bits corresponding to each fragment

4. Set ones to the calculated bits and zero for all other among 2048

FreeSolv dataset

FreeSolv: Experimental and Calculated Small Molecule Hydration Free Energies

This repository provides an issue tracker and revision control for the FreeSolv database, initially described in JCAMD (10): http://dx.doi.org/10.1007/s10822-014-9747-x. If you find any issues, please raise an issue in the issue tracker or file a pull request!

Releases are automatically assigned unique DOIs via Zenodo. Latest release: DOI 10.5281/zenodo.1161245

- contains both experimental and calculated hydration energies for over 600 molecules

https://github.com/MobleyLab/FreeSolv